

SHASIN MADUSHAN

Data Science Undergraduate

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PROFESSIONAL SUMMARY

Detail-oriented Data Science Undergraduate and aspiring Data Engineer with proven expertise in building end-to-end data pipelines, designing relational data models, and ingesting complex data from diverse sources. Highly proficient in SQL and Python, with a strong track record of developing interactive business intelligence dashboards and monitoring pipeline health to support data-driven commercial decisions. Known for a strong problem-solving knack and bridging the gap between raw data and stakeholder requirements.

CORE COMPETENCIES

Data Engineering & Pipelines: Data Ingestion, Pipeline Health Monitoring, Apache Kafka, ETL/ELT (SSIS), AWS Cloud.

Data Modeling & Databases: Relational Database Design, Star Schema, PostgreSQL, Microsoft SQL Server, MySQL, MongoDB.

Programming & Logic: Python, SQL, Java, Node.js, JavaScript, Object-Oriented Programming (OOP).

Dashboards & BI: Power BI, SQL Server Analysis Services (SSAS) Cubes, Data Visualization.

EDUCATION

G.C.E. Advanced Level (A/L) (2022(2023))

St. Annes College , Kurunagala ,Sri Lanka

- Chemistry – B
- Combined Mathematics - C
- Physics - C

Z-Score - 1.0533

BSc (Hons) in Information Technology, Specializing in Data Science

2023 – Present

Sri Lanka Institute of Information Technology (SLIIT)

- Current GPA:** 3.67
- Awards:** Dean's List Award for Academic Excellence; HackerRank SQL Gold Badge (Advanced query optimization).
- Continuous Learning:** Pursuing Harvard University's CS50P (Python) & KodeKloud's 50 Days of AWS Challenge (Day 28/50).

TECHNICAL INGESTION & PIPELINE PROJECTS

Olist E-Commerce Real-Time Data Pipeline

Solo Project

End-to-End Cloud Data Pipeline (ELT Architecture) | *Python*,
Apache Kafka, *AWS RDS (PostgreSQL)*, *SQL*

- Real-Time Data Streaming:** Engineered a custom Python/Kafka producer to extract and stream over 300,000 raw e-commerce records (customers, orders, payments) to a Confluent Cloud cluster, simulating a high-throughput transaction environment.

- **Optimized Cloud Ingestion:** Prevented network bottlenecking and API rate limits by programming a Kafka Consumer to hold live streams in memory, executing targeted bulk-inserts (5,000 records/batch) directly into an AWS PostgreSQL Data Warehouse.
- **Data Modeling & Analytics:** Architected a Bronze-to-Gold data model by writing cloud-native SQL to clean, join, and aggregate disconnected tables into a 99,000+ row analytics view, leveraging Pandas and Seaborn to automate business intelligence dashboards.
- **Architecture Roadmap:** Upgrading pipeline to V2.0 by decoupling transformations with a distributed PySpark cluster and implementing Apache Airflow for DAG orchestration and automated scheduling.

Banking Data Warehouse & Business Intelligence System

Academic Project

Target Stack: SQL Server, SSIS, SSAS, Power BI, Star Schema

- **Data Warehousing & Modeling:** Designed and implemented a banking data warehouse using Star Schema architecture with centralized fact and dimension tables for analytical processing.
- **ETL & Data Integration:** Developed SSIS ETL workflows to extract, cleanse, transform, and load transactional data from staging into the data warehouse.
- **Historical Data Management:** Implemented SCD Type 1 and Type 2 techniques to manage changing customer and transactional data efficiently.
- **Business Intelligence & Analytics:** Built SSAS OLAP cubes and interactive Power BI dashboards to analyze KPIs, transaction trends, and business insights.

Smart Campus Operation Hub

Full-Stack Architecture

Target Stack: Java, Spring Boot, React, MySQL

- **Problem Solving & Logic:** Developed a strict chronological conflict-validation engine for resource booking to prevent overlapping schedules and data entry errors.
- **Backend Development:** Built a dynamic state-machine to ingest real-time maintenance ticketing data and automatically dispatch technicians based on priority rules.

Interactive Data Engineering Portfolio

Frontend Architecture

Target Stack: HTML, CSS, Vanilla JavaScript, DOM Logic

- **Web Development:** Deployed a modern, high-performance portfolio website engineered to showcase large-scale data engineering pipelines and cloud credentials.
- **System Design:** Implemented lightweight DOM logic and custom keyframe animations to create a responsive, fluid user interface.

CERTIFICATIONS

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- Data Engineering Fundamentals – KodeKloud
 - Event Streaming with Apache Kafka – KodeKloud
 - MongoDB Data Modeling Path – MongoDB University
 - AWS Cloud Crash Course – KodeKloud